

# List of Tables



|     |                                                                                                                                                                                        |     |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2.1 | The unweighted initial abundances by mass of our models, unless otherwise stated. . . . .                                                                                              | 18  |
| 3.1 | The five major stages of the evolution of our binary. . . . .                                                                                                                          | 54  |
| 3.2 | The key evolutionary parameters of both stars and the binary configuration at five critical points: the initial parameters, and the beginning and ends of both epochs of RLOF. . . . . | 64  |
| 4.1 | The populations under study. . . . .                                                                                                                                                   | 73  |
| 4.2 | The systems we use in our study. . . . .                                                                                                                                               | 82  |
| 5.1 | Search parameters for the six close-contact binaries compiled by <a href="#">Abdul-Masih et al. (2022)</a> , as used in our fitting to the BPASS v2.2.1 grid. . . . .                  | 98  |
| 5.2 | Matches from the HOKI ( <a href="#">Stevance et al., 2020</a> ) software. . . . .                                                                                                      | 101 |
| 5.3 | Matches from the TUI code. . . . .                                                                                                                                                     | 102 |
| 5.4 | The mean post-secondary-supernova parameters for the bound remnants of our binary models. . . . .                                                                                      | 105 |
| 5.5 | The matches in primary mass / primary radius to our overcontact models. . . . .                                                                                                        | 109 |
| 5.6 | The observed vs. matched parameters for each of the 14 overcontact matches we find. . . . .                                                                                            | 113 |
| 5.7 | The match results with associated mean ages at match. . . . .                                                                                                                          | 114 |
| 6.1 | The runtimes for a grid of STARS models on a 72-core cluster, assuming each model takes 20 min/40 min/60 min respectively. . . . .                                                     | 123 |
| A.1 | The mass loss rate constants for interpolation around the bi-stability jump. . . . .                                                                                                   | 142 |
| B.1 | The code versions we use in this thesis. *TUI was described in earlier papers, but <a href="#">Ghodla et al. (2023)</a> was the first to name it. . . . .                              | 146 |
| B.2 | Our general initial conditions. . . . .                                                                                                                                                | 147 |
| D.1 | Matches from the TUI code. . . . .                                                                                                                                                     | 155 |
| E.1 | The remnants generated by each model from the BPASS release. . . . .                                                                                                                   | 178 |

